Unit 5 – Trigonometric & Exponential Derivatives.

* Consider $f(x) = \sin(x)$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

compound angle.

$$= \lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)[\cos(h) - 1] + \sin(h)\cos(x)}{h}$$

No Need to know!

$$= \lim_{h \to 0} \frac{\sin(x)[\cos(h) - 1]}{h} + \lim_{h \to 0} \frac{\sin(h)\cos(x)}{h}$$

$$= \sin(x) \cdot \underbrace{\lim_{h \to 0} \frac{\cos(h) - 1}{h}} + \cos(x) \cdot \underbrace{\lim_{h \to 0} \frac{\sin(h)}{h}}$$

$$= \cancel{\sin(x) \cdot 0} + \cos(x) \cdot 1$$ * see DESMOS

$$= \cos(x)$$

$\therefore$ $f(x) = \sin(x)$

$f'(x) = \cos(x)$

* Consider $f(x) = \cos(x)$

$$\cos(x) = \overbrace{\sin\left(\frac{\pi}{2} - x\right)}$$

chain Rule

$$\frac{d}{dx} \sin\overset{\theta}{\left(\frac{\pi}{2} - x\right)} = \cos\overset{\theta}{\left(\frac{\pi}{2} - x\right)}(-1)$$

$$= -\cos\left(\frac{\pi}{2} - x\right)$$

corelated

$$= -\sin(x)$$

$\therefore \quad f(x) = \cos(x)$

$f'(x) = -\sin(x)$

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* consider $f(x) = \tan(x)$

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$f'(x) = \frac{[\sin(x)]'\cos(x) - \sin(x)[\cos(x)]'}{\cos^2(x)}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)}$$

$$= \frac{1}{\cos^2(x)}$$

$$= \sec^2(x)$$

$\therefore \quad f(x) = \tan(x)$

$f'(x) = \sec^2(x)$

The Reciprocal Functions

* Consider $f(x) = \csc(x)$

$$\csc(x) = \frac{1}{\sin(x)}$$

reciprocal

$$f'(x) = \frac{0 - \cos(x)}{\sin^2(x)}$$

$$= \frac{-\cos(x)}{\sin^2(x)}$$

$$\underbrace{\frac{\cos(x)}{\sin(x)}} \cdot \underbrace{\frac{1}{\sin(x)}}$$

$$= -\cot(x)\csc(x)$$

$\therefore$ $f(x) = \csc(x)$

$f'(x) = -\cot(x) \cdot \csc(x)$

* Consider $f(x) = \sec(x)$

$$\sec(x) = \frac{1}{\cos(x)}$$

$$f'(x) = \frac{0 + \sin(x)}{\cos^2(x)} = \underline{\frac{\sin(x)}{\cos(x)}} \cdot \frac{1}{\cos(x)}$$

$$= \frac{\sin(x)}{\cos^2(x)}$$

$$= \underline{\sec(x)}\,\underline{\tan(x)}$$

$\therefore$ $f(x) = \sec(x)$

$f'(x) = \sec(x)\tan(x)$

* Consider $f(x) = \cot(x)$

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$f'(x) = \frac{-\sin(x)\sin(x) - \cos(x)\cdot\cos(x)}{\sin^2(x)}$$

$$= \frac{-(\sin^2(x) + \cos^2(x))}{\sin^2(x)}$$

$$= \frac{-1}{\sin^2(x)}$$

$$= -\csc^2(x)$$

$\therefore$ $f(x) = \cot(x)$

$f'(x) = -\csc^2(x)$

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Example: Determine y' for each of the following.

a) $y = 3\sin(x)$

$y' = 3\cos(x)$

b) $y = \sin(x)\cos(x)$

$y' = \sin(x)'\cos(x) + \sin(x)\cos(x)'$

$y' = \cos^2(x) - \sin^2(x)$ } double angle

$y' = \cos(2x)$

c) $y = 3\pi\sin(\theta) - \frac{\pi}{2}\cos(\theta)$

$y' = 3\pi\cos(\theta) + \frac{\pi}{2}\sin(\theta)$

$y' = \frac{\pi}{2}\left(6\cos(\theta) + \sin(\theta)\right)$

$\frac{\sin^2(x)}{\cos(x)}$ Q.R.

d) $y = \sin(x)\tan(x)$

$y' = \sin(x)'\tan(x) + \sin(x)\tan(x)'$

$y' = \cos(x)\tan(x) + \sin(x)\sec^2(x)$

$y' = \sin(x) + \sin(x)\sec^2(x)$

$1 + \tan^2(x) = \sec^2(x)$

$y' = \sin(x)\left(1 + \sec^2(x)\right)$

$\sin(x) = 0$

$1 + \sec^2(x) = 0$

$\sec^2 x = -1$.

The Chain Rule.

$f(x) = \sin[g(x)]$

$\sec(g(x))$

$f'(x) = \cos[g(x)] \cdot g'(x)$ $\rightarrow \sec(g(x)) \cdot \tan(g(x)) \cdot g'(x)$

$\therefore$ $f(x) = \sin[g(x)]$

$f'(x) = \cos[g(x)] \cdot g'(x)$

Example: Differentiate the following.

a) $y = \sin(4x)$

$y' = 4 \cdot \boxed{\cos(4x)}$ $\rightarrow$ $\cos(4x)4.$

b) $y = \sin(x^2+3)$

$y' = 2x \cdot \cos(x^2+3)$

$\neq \cos(x^2+3)(2x)$

c) $f(x) = \cos[\sin(2x^2-3)]$

$f'(x) = -\sin[\sin(2x^2-3)] \cdot \cos(2x^2-3) \cdot (4x)$

$f'(x) = -4x \sin[\sin(2x^2-3)] \cdot \cos(2x^2-3)$

d) $y = \cos^2(3x^2+6x+4) = [\cos(3x^2+6x+4)]^2$

$y' = 2\cos'(3x^2+6x+4) \cdot ([-\sin(3x^2+6x+4)](6x+6))$

$y' = -6(x+1)\sin[2(3x^2+6x+4)]$ double angle.

Example: Determine the equation of the tangent line and the normal line of $f(x) = \sec(x) + \tan(x)$ at $x = \frac{\pi}{4}$.

Solution:

$$f'(x) = \sec(x)\tan(x) + \sec^2(x)$$

$$f'\left(\frac{\pi}{4}\right) = \sec\left(\frac{\pi}{4}\right)\tan\left(\frac{\pi}{4}\right) + \sec^2\left(\frac{\pi}{4}\right)$$

$$f'\left(\frac{\pi}{4}\right) = \sqrt{2} + 2$$ exact

Now, $f\left(\frac{\pi}{4}\right) = \sec\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right)$

$$f\left(\frac{\pi}{4}\right) = \sqrt{2} + 1$$

$$\therefore \; y - y_1 = m_t(x - x_1)$$

$$\left.y - (\sqrt{2}+1) = (\sqrt{2}+2)\left(x - \frac{\pi}{4}\right)\right\} \text{ equation of tangent line!}$$

$$\therefore \; y - y_1 = m_\perp(x - x_1)$$

$$\left.y - (\sqrt{2}+1) = \left(\frac{-1}{\sqrt{2}+2}\right)\left(x - \frac{\pi}{4}\right)\right\} \text{ equation of normal line!}$$